Worksheet 3.6 — Trigonometric Functions and Equations

Mathematics: Analysis & Approaches SL (IB) · Unit 3: Geometry and Trigonometry

The Big Ideas

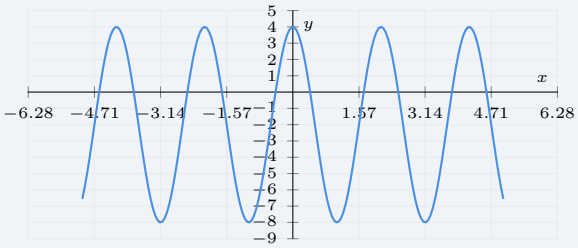
Solving trig equations combines the unit circle's symmetry with ordinary algebra — including factoring quadratics in $\sin x$ or $\cos x$.

- $y = a\sin(b(x-c)) + d$: amplitude $|a|$, period $\dfrac{2\pi}{b}$, midline $y = d$.
- Quadratic-in-trig equations factor exactly like ordinary quadratics after substituting $u = \sin x$ or $u = \cos x$.

Worked Examples

Example 1 — State transformation features

State the amplitude, period, and midline of $y = 6\cos(3x) - 2$.



Solution. Amplitude 6, period $\dfrac{2\pi}{3}$, midline $y = -2$.

Conclusion: amplitude 6, period $2\pi/3$, midline $y = -2$.

Example 2 — Solve a linear trig equation

Solve $\cos x = -0.4$ for $0 \le x \le 2\pi$.

Solution. Reference angle $\cos^{-1}(0.4) \approx 1.159$. Q2: $x \approx \pi - 1.159 \approx 1.983$. Q3: $x \approx \pi + 1.159 \approx 4.301$.

Conclusion: $x \approx 1.983, 4.301$.

Example 3 — Quadratic in sine

Solve $2\sin^2 x + \sin x - 1 = 0$ for $0 \le x \le 2\pi$.

Solution. $(2\sin x - 1)(\sin x + 1) = 0$. $\sin x = \frac{1}{2} \Rightarrow x = \frac{\pi}{6}, \frac{5\pi}{6}$. $\sin x = -1 \Rightarrow x = \frac{3\pi}{2}$.

Conclusion: $x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$.

Example 4 — Identity substitution first

Solve $2\cos^2 x - \sin x - 1 = 0$ for $0 \le x \le 2\pi$.

Solution. $2(1 - \sin^2 x) - \sin x - 1 = 0 \Rightarrow -2\sin^2 x - \sin x + 1 = 0 \Rightarrow 2\sin^2 x + \sin x - 1 = 0 \Rightarrow (2\sin x - 1)(\sin x + 1) = 0$.

Conclusion: $x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$ (same equation as Example 3, after substitution).

Example 5 — Extended domain (more solutions)

Solve $\sin x = 0.5$ for $0 \le x \le 4\pi$.

Solution. In $[0, 2\pi]$: $x = \frac{\pi}{6}, \frac{5\pi}{6}$. Add 2π for the second cycle: $x = \frac{13\pi}{6}, \frac{17\pi}{6}$.

Conclusion: $x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{13\pi}{6}, \frac{17\pi}{6}$.

Example 6 — Quadratic-in-cosine requiring the quadratic formula

Solve $3\cos^2 x - \cos x - 1 = 0$ for $0 \le x \le 2\pi$ (this doesn't factor nicely).

Solution. Let $u = \cos x$: $u = \dfrac{1 \pm \sqrt{1+12}}{6} = \dfrac{1 \pm \sqrt{13}}{6}$, giving $u \approx 0.768$ or $u \approx -0.434$.

Conclusion: $x \approx 0.696,\ 2.009,\ 4.275,\ 5.588$ (four solutions, from the two cosine values each giving two angles).

Example 7 — Factoring out a common trig factor

Solve $2\sin x \cos x - \sin x = 0$ for $0 \le x \le 2\pi$.

Solution. Factor: $\sin x(2\cos x - 1) = 0$. $\sin x = 0 \Rightarrow x = 0, \pi, 2\pi$. $\cos x = \frac{1}{2} \Rightarrow x = \frac{\pi}{3}, \frac{5\pi}{3}$.

Conclusion: $x = 0, \frac{\pi}{3}, \pi, \frac{5\pi}{3}, 2\pi$.

Example 8 — Finding a maximum and when it first occurs

A wave's height is $h(t) = 5\sin(2t) + 3$; find the maximum height and the first time $t > 0$ it's reached.

Solution. Maximum of sin is 1, giving $h = 5 + 3 = 8$, at $2t = \frac{\pi}{2}$.

Conclusion: max height 8, first reached at $t = \frac{\pi}{4}$.

Example 9 — Application — bridge clearance

A bridge's vertical clearance is $C(t) = 2\cos\left(\dfrac{\pi}{6}t\right) + 5$ m, for $t \in [0, 24)$ hours. Find all times the clearance equals 4 m.

Solution. $2\cos(\pi t/6) = -1 \Rightarrow \cos(\pi t/6) = -0.5 \Rightarrow \pi t/6 = \frac{2\pi}{3}$ or $\frac{4\pi}{3}$ (period is 12 h, so this repeats once more in $[0, 24)$).

Conclusion: $t = 4,\ 8,\ 16,\ 20$ hours.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

State the amplitude, period, and midline of $y = 4\sin(2x) + 3$.

Question 2

Solve $\sin x = 0.65$ for $0 \leq x \leq 2\pi$.

Question 3

Solve $2\cos^2 x - \cos x - 1 = 0$ for $0 \leq x \leq 2\pi$.

Question 4 — Challenge

Solve $2\sin^2 x - \cos x - 1 = 0$ for $0 \leq x \leq 2\pi$ (substitute $\sin^2 x = 1 - \cos^2 x$ first).

Question 5

A tide's depth is $D(t) = 3\sin\left(\frac{\pi}{6}t\right) + 7$ metres; find the first two times in $0 \le t \le 12$ hours the depth is 8.5 m.

Question 6

Solve $\tan x = -1$ for $0 \le x \le 2\pi$.

Question 7

Solve $2\sin^2 x + 3\sin x + 1 = 0$ for $0 \le x \le 2\pi$.

Question 8 — Challenge

A Ferris wheel's height is $h(t) = 10\sin\left(\frac{\pi}{4}t\right) + 12$ m; find every time in the first 8 minutes it reaches height 17 m.

Question 9

Solve $\cos x = 0.3$ for $0 \le x \le 4\pi$.

Question 10

Solve $2\sin^2 x + \sin x = 0$ for $0 \le x \le 2\pi$.

Question 11 — Challenge

Solve $2\sin^2 x - 3\sin x - 2 = 0$ for $0 \le x \le 2\pi$ (reject any invalid value of $\sin x$).

Question 12 — Challenge

Solve $2\sin^2 x + \sin x - 2 = 0$ for $0 \le x \le 2\pi$ (does not factor nicely — use the quadratic formula).

Question 13 — Challenge

A bridge's clearance is $C(t) = 2\cos\left(\frac{\pi}{6}t\right) + 5$ m for $t \in [0, 24)$; find all times the clearance equals 3.5 m (nearest 0.01 h).

Question Recap & Answer Key

Question Recap

1. State the amplitude, period, and midline of $y = 4\sin(2x) + 3$.
2. Solve $\sin x = 0.65$ for $0 \le x \le 2\pi$.
3. Solve $2\cos^2 x - \cos x - 1 = 0$ for $0 \le x \le 2\pi$.
4. Solve $2\sin^2 x - \cos x - 1 = 0$ for $0 \le x \le 2\pi$ (substitute $\sin^2 x = 1 - \cos^2 x$ first).
5. A tide's depth is $D(t) = 3\sin\left(\frac{\pi}{6}t\right) + 7$ metres; find the first two times in $0 \le t \le 12$ hours the depth is 8.5 m.
6. Solve $\tan x = -1$ for $0 \le x \le 2\pi$.
7. Solve $2\sin^2 x + 3\sin x + 1 = 0$ for $0 \le x \le 2\pi$.
8. A Ferris wheel's height is $h(t) = 10\sin\left(\frac{\pi}{4}t\right) + 12$ m; find every time in the first 8 minutes it reaches height 17 m.
9. Solve $\cos x = 0.3$ for $0 \le x \le 4\pi$.
10. Solve $2\sin^2 x + \sin x = 0$ for $0 \le x \le 2\pi$.
11. Solve $2\sin^2 x - 3\sin x - 2 = 0$ for $0 \le x \le 2\pi$ (reject any invalid value of $\sin x$).
12. Solve $2\sin^2 x + \sin x - 2 = 0$ for $0 \le x \le 2\pi$ (does not factor nicely — use the quadratic formula).
13. A bridge's clearance is $C(t) = 2\cos\left(\frac{\pi}{6}t\right) + 5$ m for $t \in [0, 24)$; find all times the clearance equals 3.5 m (nearest 0.01 h).

✓ Answer Key

1. amplitude 4, period π, midline $y = 3$
2. reference $\sin^{-1}(0.65) \approx 0.708$; $x \approx 0.708,\ \pi - 0.708 \approx 2.434$
3. $(2\cos x + 1)(\cos x - 1) = 0$: $\cos x = -\frac{1}{2} \Rightarrow x = \frac{2\pi}{3}, \frac{4\pi}{3}$; $\cos x = 1 \Rightarrow x = 0, 2\pi$
4. $2(1 - \cos^2 x) - \cos x - 1 = 0 \Rightarrow -2\cos^2 x - \cos x + 1 = 0 \Rightarrow 2\cos^2 x + \cos x - 1 = 0 \Rightarrow (2\cos x - 1)(\cos x + 1) = 0$: $\cos x = \frac{1}{2} \Rightarrow x = \frac{\pi}{3}, \frac{5\pi}{3}$; $\cos x = -1 \Rightarrow x = \pi$
5. $\sin(\pi t/6) = 0.5 \Rightarrow \pi t/6 = \pi/6$ or $5\pi/6 \Rightarrow t = 1$ or $t = 5$ hours
6. Q2 and Q4: $x = \frac{3\pi}{4}, \frac{7\pi}{4}$
7. $(2\sin x + 1)(\sin x + 1) = 0$: $\sin x = -\frac{1}{2} \Rightarrow x = \frac{7\pi}{6}, \frac{11\pi}{6}$; $\sin x = -1 \Rightarrow x = \frac{3\pi}{2}$
8. $10\sin(\pi t/4) = 5 \Rightarrow \sin(\pi t/4) = 0.5 \Rightarrow \pi t/4 = \pi/6$ or $5\pi/6 \Rightarrow t = \frac{2}{3}$ or $\frac{10}{3}$ min (period 8, so no repeats within 8 min beyond these)
9. reference ≈ 1.266; solutions $\approx 1.266,\ 5.017,\ 7.549,\ 11.300$
10. factor $\sin x(2\sin x + 1) = 0$: $\sin x = 0 \Rightarrow x = 0, \pi, 2\pi$; $\sin x = -\frac{1}{2} \Rightarrow x = \frac{7\pi}{6}, \frac{11\pi}{6}$
11. $(2\sin x + 1)(\sin x - 2) = 0$; $\sin x = 2$ rejected (impossible); $\sin x = -\frac{1}{2} \Rightarrow x = \frac{7\pi}{6}, \frac{11\pi}{6}$
12. $u = \dfrac{-1 \pm \sqrt{17}}{4}$: $u \approx 0.781$ (valid) or $u \approx -1.281$ (rejected); $x \approx 0.896,\ 2.246$
13. $\cos(\pi t/6) = -0.75 \Rightarrow \pi t/6 \approx 2.419$ or 3.864 (period 12h): $t \approx 4.62,\ 7.38,\ 16.62,\ 19.38$